Python Activity-Based Learning
35. Text Compression Using Repeated Characters

Simple student-style assignment | Easy English | Step-by-step solution

1. Problem Statement

Write a program to shorten a string by writing each repeated character followed by the number of times it appears continuously. For example, aaabb becomes a3b2.

2. Objectives

• Read a string.

• Count continuous repeated characters.

• Build and display a compressed string.

3. Assumptions

• The input contains letters only for this simple activity.

• The input is not empty.

4. Algorithm (Step by Step)

1. Start.

2. Read a string.

3. Set the first character as the current character and count it as one.

4. Check the next characters one by one.

5. If the next character is the same, increase the count.

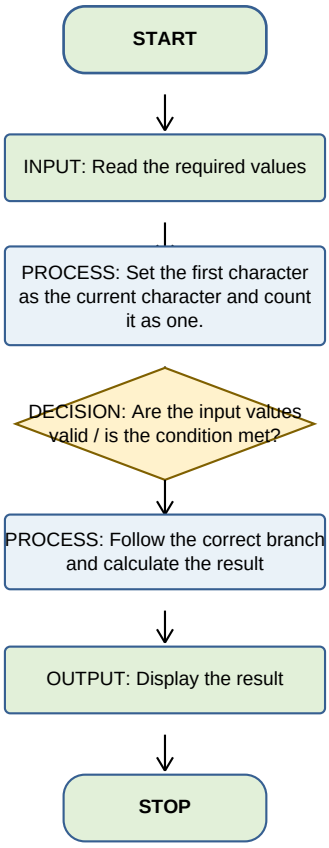
6. If it is different, add the old character and count to the result, then start counting the new character.

7. Add the last character group to the result.

8. Display the compressed string and stop.

5. Flowchart

Follow the arrows from START to STOP. The diamond shows a decision point.



6. Pseudocode

```
START
INPUT text
result = empty string
current = first character; count = 1
FOR each next character
    IF character = current THEN count = count + 1
    ELSE add current and count to result; current = character; count = 1
ADD final current and count to result
PRINT result
STOP
```

7. Python Source Code

```
text = input("Enter text: ")

if text == "":
    print("Please enter some text")
else:
    result = ""
    current = text[0]
    count = 1

    for ch in text[1:]:
        if ch == current:
            count += 1
        else:
            result += current + str(count)
            current = ch
            count = 1

    result += current + str(count)
    print("Compressed text:", result)
```

8. Test Cases

Input / Test	Expected result	What it checks
aaabb	a3b2	Two groups

Input / Test	Expected result	What it checks
abcd	a1b1c1d1	No repeated neighbours
xxxx	x4	One repeated group

9. Sample Output

```
Enter text: aaabb
Compressed text: a3b2
```

10. Conclusion

This program uses string processing and a loop to count consecutive repeated characters.